F23PP466-VS
VARIABLE SPEED DRIVE

SYSTEM INTERCONNECTION DIAGRAM DRG. No. 2000-EL005-B02-10002

11,000V

RD WH BU

FCF 160A 160A 160A

QAC 400A

BCT1 50 / 1A 2.5VA 5P20

BCT2 50 / 1A 2.5VA 5P20

BCT3 50 / 1A 2.5VA 5P20

SWITCHBOARD EARTH BAR

NOTE 3

SUB FLOOR DROP BOX

BCN 50 / 1A 1.5VA 10P10

BAD

PFV1 RD RD RD

QCE

EARTH BAR

TRANSFORMER EARTH BAR

L1 L2 L3

LINE SIDE INCOMING SUPPLY SHEET 6

X4-1 X4-2 X4-3 X4-4 X4-5 X4-6 X4-7 X4-8 X4-11 X4-12 X4-13

2000-EL003-F16-10011 SHEET 14

OUTGOING MOTOR SUPPLY SHEET 6

T1 T2 T3

CABLE SCREEN EARTHED

M 3~

F23-PP-466
FINAL TAILS PUMP 1
3,300V
800 kW

MOTOR SPACE HEATER CIRCUIT SHEET 6

X2 54 53

F23-PP-466
MOTOR HEATER TERMINAL BOX
2000-ME063B-F14-10013

F23PP466HT-L 1 (H1) 0.3kW ANTI-CONDENSATION HEATER

F23PP466HT-N 3 (H2)

A80 (PART OF)
RTD TEMPERATURE MONITOR RELAY
SHEET 6

X81

F23-PP-466
MOTOR RTD TERMINAL BOX
2000-ME063B-F14-10013

PT100 RTD INPUT CH1	F23PP466A-RD	MOTOR WINDING RTD (RD PHASE) (TD1)	
	F23PP466A-WH		
	F23PP466A-BK		
PT100 RTD INPUT CH2	F23PP466B-RD	MOTOR WINDING RTD (WH PHASE) (TD3)	
	F23PP466B-WH		
	F23PP466B-BK		
PT100 RTD INPUT CH3	F23PP466C-RD	MOTOR WINDING RTD (BU PHASE) (TD5)	
	F23PP466C-WH		
	F23PP466C-BK		
PT100 RTD INPUT CH4	F23PP466D-RD	MOTOR BEARING RTD (DRIVE END) (BD10)	
	F23PP466D-WH		
	F23PP466D-BK		
PT100 RTD INPUT CH5	F23PP466E-RD	MOTOR BEARING RTD (NON DRIVE END) (BD20)	
	F23PP466E-WH		
	F23PP466E-BK		
PT100 RTD INPUT CH6			
PT100 RTD INPUT CH7			
PT100 RTD INPUT CH8			

FIELD CABLE TERMINALS

F23-SW-016 11kV FEEDER START/STOP CONTROL 2000-EL003-F16-10011 SHEET 7 3A & 3B

X3 1139 3 — X2 67

X3 1383 17 — X2 68 K41 K41

MV CONTACTOR ENABLE CIRCUIT SHEET 6

F23-SW-016 11kV FEEDER VSD HEALTHY 2000-EL003-F16-10011 SHEET 10 4A

X2 1179 4 — X2 63 K41 K41 K41

X2 1199 25 — X2 64

MV INPUT ENABLE CIRCUIT SHEET 6

F23-SW-016 11kV FEEDER VSD TRIP 2000-EL003-F16-10011 SHEET 10 3A

X2 1219 3 — X2 65

X2 1239 24 — X2 66 K41 K41 K41

MV INPUT TRIP CIRCUIT SHEET 6

F23-SW-016 11kV FEEDER ARC FLASH TRIP 2000-EL003-F16-10011 SHEET 8 5A & 5D

X2 1259 56 — X2 70

X2 1279 44 — X2 69

A30 (PART OF) X2 15 TRIP X2 16

ARC FLASH DETECTION TRIP SIGNAL SHEET 6

F23-DB-101
2000-F23-TLD-EL-10054

RD WH BU BK — 415VAC

R3 W3 B3 — 25A

EARTH BAR

F23DB101-R3 F23DB101-W3 F23DB101-B3 F23DB101-BK3

X4 1 2 3 N PE1

VSD AUXILIARY POWER SUPPLY SHEET 6

F23-DB-201
2000-F23-TLD-EL-10101

RD BK — 240VAC

Q173 — 10A

EARTH BAR

F23DB201-Q173 (RD) F23DB201-Q173 (BK)

X4 5 6 PE2

VSD UPS POWER SUPPLY SHEET 6

TO FINAL TAILS PUMPS
415V SWITCHROOM
F23-COM-001 COMMUNICATION RACK
MODBUS ETHERNET SWITCH
F23-ETH-001
2000-F23-TER-IC-10003

CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6

X47 ETHERNET ADAPTOR RJ45

SHEET 6

A2 (PART OF) DI-2B 9

X2 13 24VDC CONTROL SHEET 6 14

F23PP466-LCS
LOCAL CONTROL STATION

SAFE TORQUE OFF CIRCUIT SHEET 6

X86 1 2 3 4

F23PP466VS-1409 F23PP466VS-1419 F23PP466VS-1429 F23PP466VS-1439

X1 1 2 3 4 LOS

F23PP466VS-1459 F23PP466VS-1469

X1 5 6 E START

NOTES

1. WIRE NUMBER FERRULES ON TERMINAL STRIP SHALL BE PREFIXED WITH EQUIPMENT NUMBER e.g. F23PP466-1200, UNLESS OTHERWISE NOTED.
2. EACH WIRE SHALL BE IDENTIFIED AT BOTH ENDS WITH FERRULES.
3. IF THE CABLE SCREENS PASS THROUGH CORE BALANCE CT BEFORE BREAKING OUT THEN THE SCREEN EARTHS MUST BE BOUGHT BACK THROUGH THE CORE BALANCE CT AND TERMINATED TO EARTH.

LEGEND

- 11kV SWITCHBOARD TERMINAL
- VSD TERMINAL
- FIELD TERMINAL
- DCS OR I/O TERMINAL
- MARSHALLING TERMINAL
- DEVICE TERMINAL

PANEL A02

2000-F23-SCM-EL-10103_Rev1

F23-SW-016 - F23-PP-466 final tails pump 1 VSD feeder - schematic diagram

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\EL\F23 - FINAL TAILINGS HANDLING & STORAGE\2000-F23-SCM-EL-10103.DWG 02OCT25 9:03AM HOTSEAT04